

Near-Live Simulations to the Rescue: Lessons Learned from Using Alternative Simulation Approaches for Evaluating New Technologies

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ABSTRACT

The near-live simulation method has advantages over other methods when testing technologies for supporting team-based work because testing can be conducted without requiring an entire team to assemble. In this case study, we describe our experiences in conducting 14 remote near-live simulation sessions to evaluate a digital checklist application used in pediatric trauma resuscitation. The remote near-live simulation sessions were complex to design and conduct because participants did not have direct access to a device that could run the checklist application and the digital checklist needed to integrate with the vital sign monitor used during resuscitations. We describe how we designed the environment for running the near-live simulations and discuss the lessons learned from conducting these simulations. We highlight three factors that need to be considered when using the near-live simulation approach: (1) filming or selecting the video of the simulated event, (2) providing participants with access to the system that is being evaluated, and (3) integrating the system being tested with other systems in the simulation scenario.

CCS CONCEPTS

• **Human-centered computing** → *Usability testing*.

KEYWORDS

near-live simulation, remote user testing, pediatric trauma resuscitation

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1 INTRODUCTION

Technologies used in safety-critical, team-based settings must be thoroughly evaluated before implementation, but testing these types of systems with users can be challenging. While researchers

often use simulation sessions to evaluate systems with users, simulating team-based events can be difficult. For each session, the researcher must assemble a team, find a mutually convenient time, and reserve the appropriate setting and equipment. Issues with simulating the event can arise if team members are missing or if the appropriate setting is not available. To overcome some of these barriers, near-live simulation sessions can be used to identify workflow and usability issues before system implementation [11]. During near-live simulation sessions, a participant watches a video of an event while using the system as if they were taking part in that event. Near-live simulation sessions are especially advantageous in testing technologies used in team-based work because the system can be evaluated without assembling an entire team to simulate the scenario.

Remote evaluation sessions can also address some of the above challenges with user testing. Conducting sessions remotely reduces the travel and time costs, allowing for increased flexibility when scheduling sessions [16]. Researchers can also more easily recruit participants from different geographic areas [7, 19]. The use of remote user-centered design and testing activities increased during the COVID-19 pandemic as researchers pivoted to online sessions [14, 23]. The increase of remote meetings during the COVID-19 pandemic has also made more people open to participating in remote usability testing [22]. As remote user research sessions become prevalent, we need to further (1) study how different types of design and evaluation sessions can be run remotely, (2) identify issues with running remote sessions, and (3) brainstorm solutions.

In this case study, we discuss our experiences from conducting 14 remote near-live simulation sessions to evaluate a new version of a digital checklist application used in an emergency medical setting of pediatric trauma resuscitation. During trauma resuscitation, an interdisciplinary team of medical providers treats a severely injured patient under time pressure. The trauma team leader uses the digital checklist to guide the team through the resuscitation. Evaluating systems used during team-based, time-critical scenarios is especially challenging in clinical settings, where researchers must recruit clinicians with demanding schedules. Unless a simulation lab is available for user testing, researchers also have to reserve clinical spaces and access hospital equipment for each session. We used near-live simulation sessions to evaluate a new version of the checklist that automatically obtains vital signs from the vital signs monitor and displays the values on the checklist interface. Conducting the sessions remotely allowed us to evaluate this new version with trauma team leaders from two different hospitals. We discuss our lessons learned from conducting these near-live simulation

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sessions, focusing on three factors that need to be considered when designing near-live simulation sessions: (1) filming or selecting the video of the event, (2) providing participants with access to the system that is being evaluated, and (3) integrating the system being tested with other systems in the simulation scenario. These lessons offer insights for other researchers in academic or industry settings as they design and conduct near-live simulation sessions as part of their user testing.

2 RELATED WORK

We next describe prior work that used the near-live simulation method. We also discuss the advantages and challenges of remote usability testing, as well as tools and methods for remote sessions.

2.1 Near-Live Simulation Sessions

Near-live simulation sessions can identify issues with systems, such as poor workflow integration and unexpected user interactions, that may not be uncovered in traditional usability testing [11, 21]. Prior work has also found that users may be more critical of the system after participating in the near-live simulation session than after participating in a standard usability session [21]. The near-live simulation method has often been used to evaluate clinical decision support systems [10, 12, 15, 20, 25]. Most of these studies were conducted in person, where the researcher was co-located with the participant. Prior efforts to conduct remote simulation sessions over a video conferencing platform have been reported, but lack details about the procedure [10]. Instead of having the participant watch a video of the event, some studies had the participant interact with an actor portraying a patient [12, 20]. While an actor can simulate the patient-clinician interaction, simulating team-based events with actors is more challenging due to the increased costs and complexities associated with hiring and training multiple actors for each near-live simulation session. When conducting sessions where the participant watches videos of an event while using the system, high-quality videos that accurately depict the event are critical for identifying workflow and usability issues. In this case study, we use our lessons learned to propose suggestions for filming or selecting the videos shown to participants during near-live simulation sessions.

Prior work has mainly focused on the outcomes of near-live simulations (e.g., the usability issues). As a result, information about experimental design and procedures for these types of sessions is limited. In a previous case study, we described our experiences conducting remote user-centered design activities, which included four near-live simulation sessions to evaluate new features on the digital checklist application [14]. In this case study, we build on our prior work by discussing lessons learned from conducting a new set of 14 near-live simulation sessions. These new sessions were more complex to design and run for two reasons: (1) the participants did not have access to the device running the digital checklist and (2) we needed to integrate the digital checklist with the vital sign monitor in the simulation video because our evaluation focused on the effects of this integration on physicians' situation awareness of vital sign changes.

2.2 Remote Usability Testing: Advantages and Challenges

Usability testing can be done in person (researchers and participants are co-located) or remotely (researchers and participants are geographically separated), with the remote sessions conducted synchronously or asynchronously. Prior work has found that task performance is equivalent between in-person and remote synchronous testing sessions, with a similar number of usability issues identified by each method [3, 4]. Studies comparing synchronous and asynchronous remote testing methods have shown mixed results [1, 3, 22]. Some studies found that participants provided richer qualitative data and identified more issues during synchronous sessions [3, 22], while participants in a study that evaluated a hospital website identified more issues during asynchronous sessions [1]. Remote usability testing can have several advantages over in-person testing. Because the sessions can be conducted anywhere, researchers can recruit participants from harder-to-reach populations, such as people with chronic diseases or disabilities [24]. Participants can also prefer remote testing sessions, finding them more convenient than in-person sessions [4]. Conducting sessions remotely can also reduce time and travel costs for both participants and researchers [16].

Despite these advantages, barriers to remote usability testing remain. While it is critical to include participants with less technical knowledge in usability testing, these participants can struggle when required to use the technology to participate in a remote session [17]. Researchers also often have to invest additional time and resources to prepare for the remote sessions and ensure that the required technology is setup correctly [4, 14]. Several guidelines have been proposed to assist researchers in preparing for and conducting remote usability testing sessions. For example, to reduce barriers to participation in remote sessions, researchers can use technologies familiar to participants or share instructions for new technologies in advance [14, 27]. Researchers can also send participants packages with instruction guides, contact information, links to video walk-throughs, and smartphones with remote access applications preinstalled to increase participation across different levels of technical knowledge [8]. While preparing for sessions, researchers should also consider how to obtain informed consent and compensate participants digitally [22]. We next describe some of the tools and methods used to conduct remote user research.

2.3 Tools and Methods for Remote User Research

Different tools [6, 13] and methods [2, 5, 9, 18] have been created for conducting remote user research. Tools created for remote user research include a system for displaying paper prototypes on remote participants' smartphones [6] and a system for conducting Wizard of Oz studies while participants drive cars [13]. Methods for conducting remote user research without requiring specific devices or technologies have also been developed. To elicit feedback on virtual reality (VR) prototypes from hard-to-reach neurologists, pre-recorded videos of a user interacting with the VR environment were sent to the participants for watching at their convenience [2]. In the surrogate-aloud method, the participant thinks aloud while directing a surrogate, who is wearing a VR headset and sharing

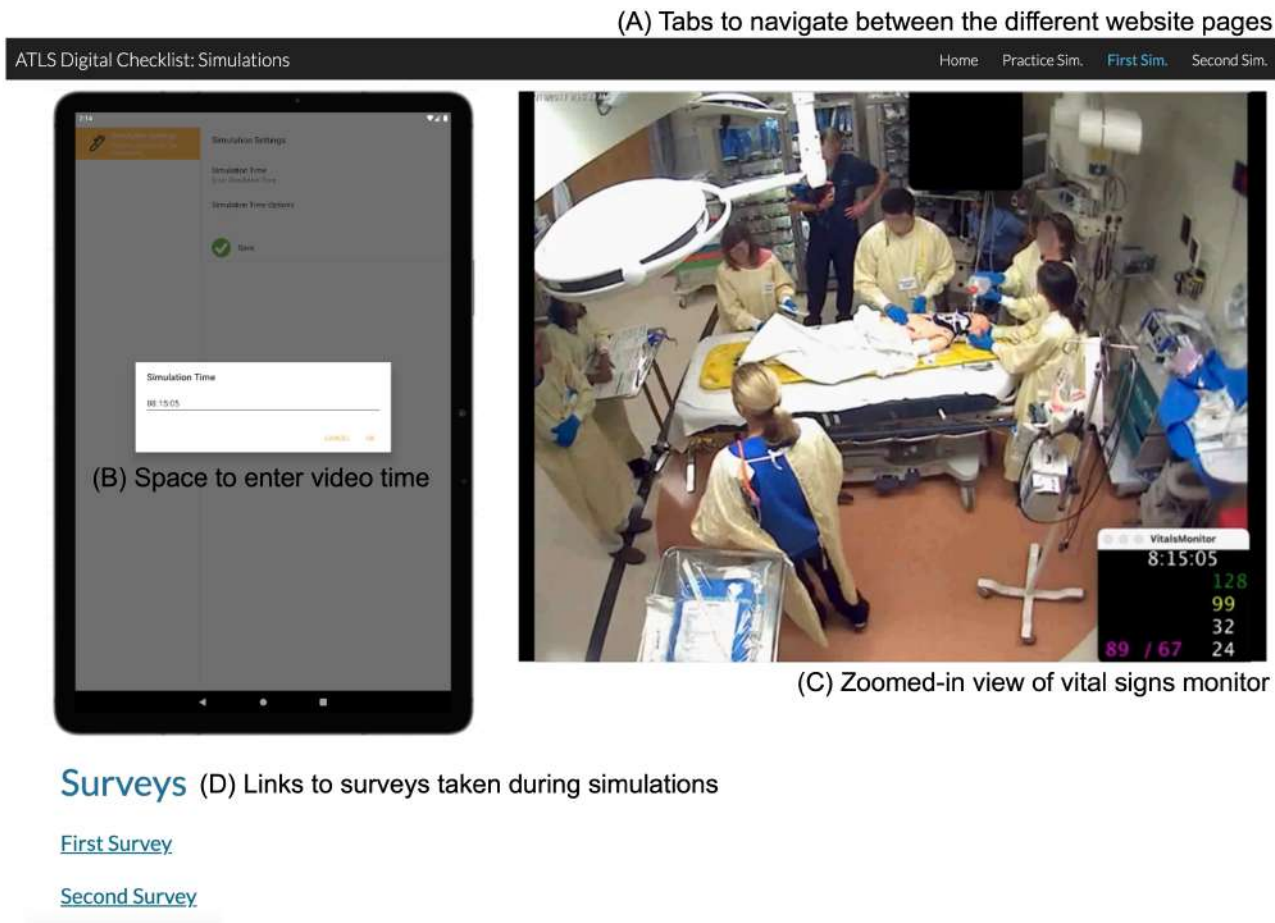


Figure 1: A screenshot of the website built for the near-live simulation sessions.

their VR viewpoint over a video conference call [9]. In another method for remote usability testing, the researcher shares their screen with the prototype and gives the participant remote control to perform tasks [18]. In this case study, we add to the repertoire of tools for remote usability testing by describing the tools we used to allow clinicians to participate in the session without needing any special devices or advanced preparation.

3 METHODS

We used near-live simulation sessions to evaluate the effects of automatically capturing and displaying vital sign values on the checklist interface on clinicians' situation awareness. In the within-subjects study, each participant watched two videos of simulated resuscitations in which the medical team evaluated and treated a "patient" (simulation mannequin) suffering from life-threatening injuries. These simulated resuscitations were previously recorded in the resuscitation room at our primary research site. While watching each video, the participant used a different version of the checklist: a version where the user had to manually document vital sign values and a version where the values were automatically captured and displayed on the checklist interface. We paused the simulations

at predetermined time points to have the participant complete a survey assessing their situation awareness.

3.1 Research Site and Participants

Our participants were 14 physicians from two different research sites with experience in leading pediatric trauma resuscitations. Both research sites were pediatric level 1 trauma centers located in the mid-Atlantic region of the United States. Participants were senior residents, fellows, and attending physicians. Senior residents are completing their general training while rotating at different hospitals. Fellows have finished their general training and are pursuing sub-specialty training. Attending physicians are permanent hospital employees with extensive clinical experience. Six participants had experience in using the digital checklist during trauma resuscitation while the other eight participants were using it for the first time during the near-live simulation sessions. All participants were compensated for their time with gift cards that we distributed electronically. The study was approved by the Institutional Review Board at our primary research site.

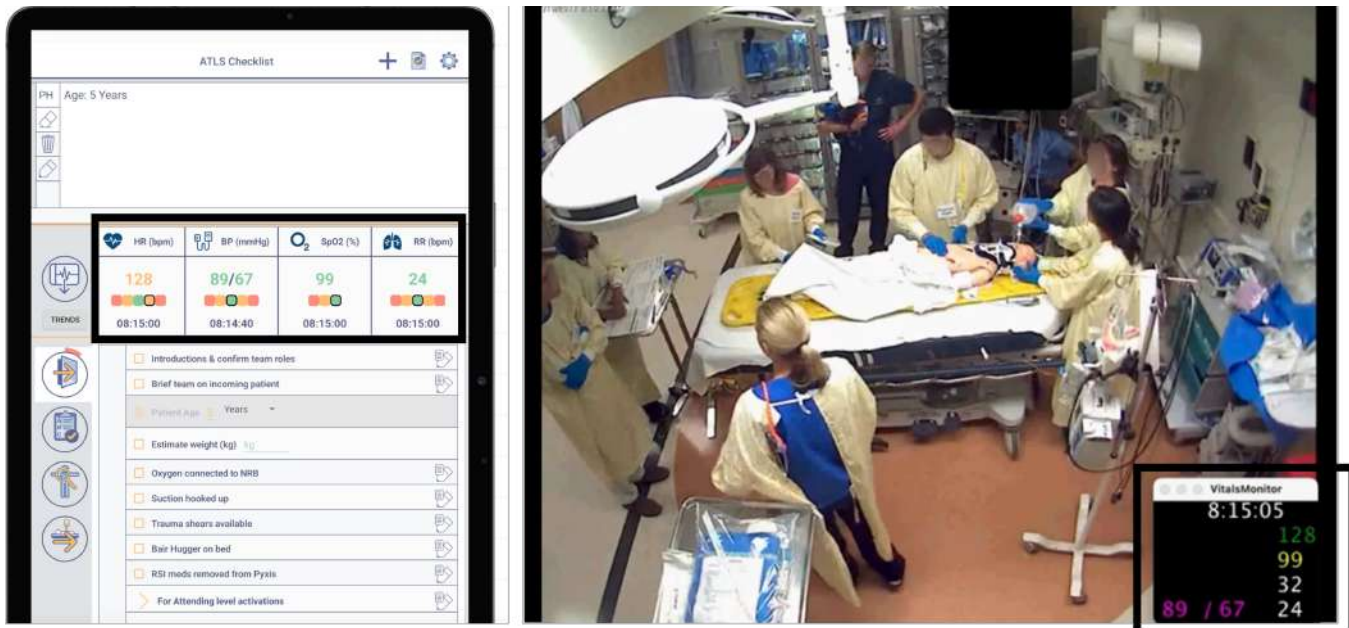


Figure 2: The automatic checklist needed to simulate the automatic capture and display of vital signs from the monitor in the video.

3.2 Technical Setup

We had three requirements for conducting our remote sessions: (1) participants do not need access to a device that could run the checklist application, (2) participants do not need to prepare in advance for the session, and (3) the automatic version of the checklist must integrate with the vital sign monitor in the simulation video. To satisfy these requirements, we created a website that served as our testing environment during the near-live simulation sessions (Figure 1). The "Home" page contained two tablets that we embedded using the Appetize service, which hosts mobile applications in a web browser. We installed a different version of the checklist (manual vs. automatic) on each tablet to instruct participants on the differences between the two versions. The website also had three "Simulation" pages: one page for the user to complete a practice simulation and two pages for the actual simulations (Figure 1.A). Each simulation page contained an embedded tablet running the digital checklist application on the left side of the screen and a video of a resuscitation on the right side of the screen. The automatic version of the checklist needed to capture vital signs from the monitor in the video and display the values on the interface. As a result, we needed to synchronize the automatic checklist with the vital sign monitor in the video (Figure 2). To accomplish this synchronization, we created a spreadsheet for each of the videos played during the practice and actual simulations. A member of the research team watched the three different videos, stopping every ten seconds to record the timestamp and vital sign values in the spreadsheet for that video. These three spreadsheets were then included in the digital checklist application. In the settings page of the application, we created a feature to input the time from the video (Figure 1.B). During the session, we would input the video

time before we started or resumed playing the video for the participant. After the video time was inputted, a timer started within the checklist application to synchronize the checklist time with the time in the video. The automatic checklist constantly updated the vital sign values based on the values recorded in the file for that video. Because the vital sign monitor was too small and difficult to view in the video, we also created a program that displayed a similar-looking vital sign monitor in the lower right corner of the video (Figure 1.C). To display vital sign values, this program used the same spreadsheets with vital sign values as the digital checklist application, further ensuring consistency between the checklist and vital sign monitor. Links to Google Forms, containing the surveys taken by the participant during the simulations, were located at the bottom of the page (Figure 1.D).

3.3 Session Protocol

The sixty-minute sessions were conducted over the video conferencing platform, Zoom. After reviewing the participant's rights, we shared our screen to explain the simulation website. We began on the home page, where we instructed the participant on how to use each checklist version and interact with the embedded tablet. After completing this overview, we shared the link to the website with the participant, asking them to open the link and share their screen with us. At this point, we also requested remote control of their screen through Zoom to help facilitate setup throughout the session. The participant then started a practice simulation where they watched a video of the resuscitation while using the checklist on the embedded tablet. Once the participant felt comfortable with the setup, we asked them to switch to the first simulation tab (Figure 1.A). We then instructed the participant that we would begin

Filming or Selecting the Simulation Video 	Questions to Consider: • What scenario to use? • What point of view should the video be filmed from? • What systems should or should not be displayed in the video?	Suggestions: • Film the video from the point of view of the team member who typically uses the system • Highlight all components of the video to participant before beginning the simulation • Try to ensure that the scenario and team's actions are standardized
Providing the Participant with Access to System Being Evaluated 	Questions to Consider: • Can the system be run on a device commonly used by participants? • Is this system easy to install and set up? • How precisely do interactions and visual effects need to be tested?	Suggestions: • Consider using a service to embed the device in the browser if: a) Most participants do not have access to the device <i>and</i> b) Need to avoid shipping and other advanced preparation <i>and</i> c) Specific interactions and color-schemes are not the main focus of testing
Integrating the System with Other Systems In the Event 	Questions to Consider: • Do other systems in the video impact the participant's system? • Does the participant's system impact other systems in the event?	Suggestions: • Use a timer to synchronize systems if other systems in the event impact the participant's system: • Allow for rapid selection of predetermined time points • Allow for entry of any timepoint in video in case of unexpected interruptions

Figure 3: Factors to consider when designing and conducting near-live simulation sessions. The two-person icon indicates that the factor applies to in-person sessions. The person-pointing-to-a-screen icon indicates that the factor applies to remote sessions.

the first simulation and pause it at two different times to have them complete surveys that assessed their situation awareness. Once the participant was ready, we selected the video start time in the settings of the checklist application (Figure 1.B) and started playing the video. At the predetermined time points, we paused the video and asked the participant to scroll to the bottom of the page, where links to the surveys (via Google Forms) were located (Figure 1.D). After the participant finished the survey, we again inputted the video time in the checklist and resumed the video. After the participant finished the first video, we repeated this process with the second simulation. At the end of the session, we debriefed the participant about their experience using the two checklist versions. All sessions were video and audio recorded after obtaining consent from participants.

4 LESSONS LEARNED: FACTORS TO CONSIDER WHEN DESIGNING NEAR-LIVE SIMULATION SESSIONS

From our near-live simulation sessions, we found several usability and workflow issues that we did not identify in our earlier design walk-through sessions, where participants explored mock-ups of the automatic checklist version while thinking aloud [26]. For example, several participants in the near-live simulation sessions did not have the cognitive bandwidth to interact with a feature that

displayed graphs illustrating changes in vital sign values. We next discuss the lessons learned from our experiences in conducting near-live simulation sessions, highlighting three factors for consideration when designing near-live simulation sessions: (1) filming or selecting the video of the event, (2) providing the participant with access to the system being evaluated, and (3) integrating the system being tested with other systems in the simulation scenario (Figure 3). The second factor applies to remote near-live simulation sessions, where the researcher and participant are separated geographically. The first and third factors apply to both in-person (researcher and participant co-located) and remote sessions.

4.1 Filming or Selecting the Simulation Video

One main decision when developing the protocol for the near-live simulation sessions is determining the video to play for participants as they use and evaluate the system. This decision will depend on several factors, including the goals of the evaluation and any specific metrics for assessing the systems or tools. Our study focused on evaluating the effects of a dynamic checklist interface on clinicians' situation awareness, prompting us to select clinical scenarios that involved life-saving interventions and frequent changes in the patient status. Our experiences and feedback from participants highlighted other factors that should be considered when filming or selecting the video for the simulations: (1) the quality of the

video (e.g., the point of view shown, clarity of the audio), (2) the standardization of the scenario and team's actions, and (3) the other systems shown in the video.

Our videos were filmed from a fixed point of view by a camera installed on the ceiling of the resuscitation room. One participant noted how they had trouble viewing everything that was happening in the video at times because team members would be standing in front of one another or blocking the camera view. To be more realistic, the participant suggested filming the videos from the future system user's point of view. In our case, the videos would be filmed from the trauma team leader's point of view because they are the team member using the checklist during the event. This filming from a user's point of view could be achieved by mounting a camera on the user's head (e.g., a GoPro). Several participants also noted how they found it hard to hear parts of the video due to poor audio quality, multiple people talking concurrently, and extraneous noises (e.g., a person coughing). By nature, the resuscitation rooms are noisy because of the many people, beeping machines, and patients in pain. In designing a near-live simulation session, we need to balance having videos that are intelligible while also trying to maintain the true nature of the event.

Because we recruited participants from two different hospitals, we had to make sure the events shown in the video followed standard procedures and were not unique to one or the other research site. The teams in both videos followed a standardized resuscitation protocol when treating the patients. The scenarios also involved injuries and required interventions that are typical for pediatric trauma centers. Participants noted how they found the case and the team's actions realistic. Although the team followed the protocol, one participant noted how the communication practices of the team differed from the communication practices at their hospital. While steps can be taken to make the event in the video more standardized, avoiding any differences between different institutions or research sites may be challenging.

When filming or selecting the video, the systems displayed in the video should also be considered. The vital signs monitor was critical to our simulations because we were evaluating how manual versus automatic approaches to capturing the vital signs influenced participants' situation awareness and experience with the checklist. However, the vital sign monitor was hard to view in our videos. To mitigate this issue, we had to create a zoomed-in view of the monitor and insert this view into the video. Extra work can be avoided by considering the systems that should be shown before filming the video and ensuring clear views of these systems. One of our participants also suggested reviewing all systems shown in the video during the introductory parts of the near-live simulation session. This participant noted how they originally thought the vital sign monitor in our video was another Zoom box.

4.2 Providing the Participant with Access to the System Being Evaluated

Researchers can have several options for providing the participant with access to the system during the simulation: (1) require the participant to have a device that can run the system, (2) send the device to the user ahead of the session, or (3) use a service to embed

the device in a virtual or online environment, like the browser. Because we were drawing from a limited pool of potential participants (trauma resuscitation physicians), we did not want to further limit their participation by requiring access to a tablet that could run the digital checklist application. We also did not want to require the participants to receive and return a device that could run the digital checklist application due to their already demanding schedules. As a result, we decided to use a service to embed the device in the browser, allowing participants to join our sessions from their own computers. We included a practice simulation at the beginning of the session to help participants become familiar with the embedded device before starting the actual simulations. Although this option worked well for our near-live simulation sessions, it had several limitations that may be more detrimental for other studies. First, because the application was embedded in a web browser instead of a tablet, some interactions were different, rendering this approach less suitable for user research that is testing a specific interaction. For example, instead of scrolling with their finger, the participant used their mouse to scroll. If we had been evaluating the time taken for participants to navigate to different parts of the application, embedding the device in the browser may not have been the best approach. This approach would also be less suitable for testing different color schemes, as some of our participants noted how the colors looked slightly different based on their web browser settings.

4.3 Integrating the System with Other Systems in the Simulation Scenario

Some near-live simulations may require integration of the system being evaluated with other systems in the simulation scenario. In some cases, other systems in the scenario can impact how participants use the system that is being evaluated. During our near-live simulations, the checklist needed to be integrated with the vital sign monitor because the automatic version of the checklist captured the monitor data and displayed the values on the checklist interface. We found that incorporating a timer within the checklist application worked well to synchronize the vital signs displayed on the checklist and the vital signs displayed on the monitor. In the checklist settings, we provided two ways for entering the time in the video: (1) buttons to select the predetermined time points when we would pause the simulation to have the participant complete a survey and (2) a space on the interface of the settings page to enter any time from the video. The buttons allowed us to quickly select the time when resuming the simulations at the predetermined time points in the video. The free text space to enter any time in the video was useful to resume the simulation after unexpected interruptions, such as the participant needing a break or the application crashing. In other studies, user interactions with the system being tested may need to impact or alter other systems in the event. This type of system integration would be challenging to simulate in near-live simulations where the participant watches a video of the simulation scenario because the videos are filmed without knowing when participants will interact with the system and what interactions they will perform.

5 CONCLUSION AND FUTURE DIRECTIONS

Near-live simulation sessions have advantages over other usability testing methods when testing technologies used during team-based work because this approach does not require the assembly of an entire team. The sessions must be thoughtfully designed, especially when they are conducted remotely. In this case study, we discussed the lessons learned from conducting 14 near-live simulation sessions, highlighting three factors that should be considered when designing the sessions. While near-live simulation sessions allow for easier testing of technologies used during team-based activities, one limitation of this method is that participants cannot interact with the team when watching the video of the event. In the future, virtual reality could serve as a tool for conducting user testing of systems in team-based events. Within the virtual reality application, the participant could move around the setting, use the device running the application, and interact with the team. One could imagine how our simulation sessions could have been conducted more realistically and easily with virtual reality. The participant could join the virtual world, which could be a resuscitation room resembling the one at their hospital. In this virtual world, the participant could have a device running the application, which would easily integrate with other systems in the scenario. While using the system, the participant could move around the room and interact with other team members, directing their work and discussing next steps. To be successfully used in conducting user research, the virtual reality technology needs to advance so that the virtual reality sessions are not testing the limitations of the virtual reality application but allow for evaluation of the system being tested.

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